

6 HW Due Apr 13

Problem 6.1. We have a plane $z = 3x - 2y$ and a surface $z = \frac{x^3}{y^2}$. Both passes through $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

1. For the plane $z = 3x - 2y$, near the point $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, please approximately write dz as an expression of dx, dy , and estimate the value of z when (x, y) changes from $x = 1, y = 1$ to $x = 1.02, y = 1.03$. Also calculate the actual value with a computer.
2. For the surface $z = \frac{x^3}{y^2}$, near the point $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, please approximately write dz as an expression of dx, dy , and estimate the value of z when (x, y) changes from $x = 1, y = 1$ to $x = 1.02, y = 1.03$. Also calculate the actual value with a computer.
3. What is the relation between the plane and the surface?

Problem 6.2. We have two surfaces $z = x^4 + 4y$ and $z = 2x + 3y^2$. They both passes through $\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$, and they intersect at a curve.

1. Find a normal vector to $z = x^4 + 4y$ at $\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$.
2. Find a normal vector to $z = 2x + 3y^2$ at $\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$.
3. Find their cross product.
4. Find a tangent vector to the intersection curve at $\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$.

Problem 6.3. For the following functions, which directions should (x, y) move, in order to increase f in the most efficient manner (steepest slope on the graph)?

1. $f(x, y) = e^{x-y}$ at the point $(x, y) = (1, 2)$.
2. $f(x, y) = \min(2x, y)$ at the point $(x, y) = (1, 2)$. (Not differentiable)
3. $f(x, y) = \sqrt{5 - x^2 - y^2}$ at the point $(x, y) = (1, 2)$. (Boundary of the domain)

Problem 6.4. Given a surface $z = f(x, y)$, we know its normal vector at $\mathbf{p}$ is $\mathbf{N} = \begin{bmatrix} D_x f \\ D_y f \\ -1 \end{bmatrix}$. The steepest

slope tangent vector at $\mathbf{p}$ is $\mathbf{U} = \begin{bmatrix} D_x f \\ D_y f \\ (D_x f)^2 + (D_y f)^2 \end{bmatrix}$.

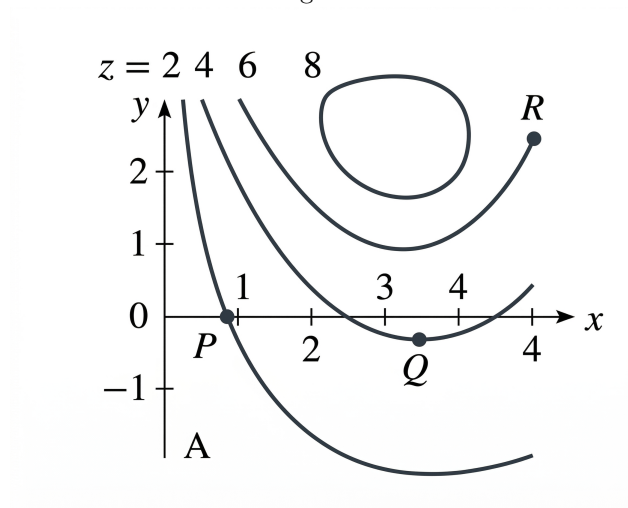
1. For the vector $\mathbf{L} = \begin{bmatrix} D_y f \\ -D_x f \\ 0 \end{bmatrix}$, show that this is a tangent vector at $\mathbf{p}$ to the level curve through $\mathbf{p}$.

2. Suppose $\nabla f \neq \mathbf{0}$, show that $(\mathbf{L}, \mathbf{N}, \mathbf{U})$ form an orthogonal right-handed basis. (You need to show that the vectors are mutually orthogonal, and that $\mathbf{U}$ is a positive multiple of $\mathbf{L} \times \mathbf{N}$.)
3. Calculate the $(\mathbf{L}, \mathbf{N}, \mathbf{U})$ frame of the surface $x^2 + y^2 - z^2 = 0$ at the point $(3, 4, 5)$.

Problem 6.5. Calculate the following.

1. $w = u^2 e^{-6v} + \cos(u^6 - 4u + 1)$, find w_{uvvv} .
2. Find the tangent plane and the normal line to the surface $\ln(\frac{x}{2y}) = z^2(x - 2y) + 3z + 3$ at the point $(4, 2, -1)$.
3. Find the level direction (the horizontal tangent vector) for the function $\theta(x, y) = \arctan(\frac{y}{x})$ at the point $(3, 4)$.
4. Give me a function $f(x, y)$ whose gradients are always normal to the hyperbolas $xy = k$.
5. Give me a function $f(x, y)$ whose gradients are always tangent to the hyperbolas $xy = k$.

Problem 6.6. The followings are level curves for a function $z = f(x, y)$.



1. Draw the directions of ∇f at the points P, Q, R .
2. Do you think the gradient is larger at $(1, 2)$ or at the point Q ?
3. On the straight line segment from P to R , where will f achieve maximum? What is the maximum value?